The Cardiac Cycle

One cardiac cycle is a single heartbeat: the period from the start of one beat to the start of the next. At rest it lasts about 0.8 seconds. Each cycle has a contraction phase, **systole**, and a relaxation phase, **diastole**.

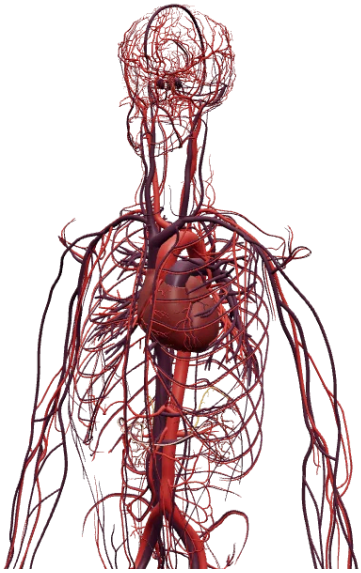


Figure 1. The heart sits between the lungs, slightly left of the midline.

Four steps

Step	What happens	Valves
1. Atrial systole	Atria contract and top up the ventricles	AV valves open
2. Ventricular systole	Ventricles contract; blood leaves through the aorta and pulmonary trunk	AV valves close ("lub"); semilunar valves open
3. Ventricular diastole	Ventricles relax; pressure falls	Semilunar valves close ("dub")
4. Filling	Blood flows from the veins through the atria into the ventricles	AV valves open

> **Heart sounds:** "lub" is the AV valves closing; "dub" is the semilunar valves closing.

Key numbers

- Resting heart rate: 60 to 100 beats per minute.
- Stroke volume: about 70 mL per beat.
- Cardiac output = heart rate × stroke volume, about 5 L per minute at rest.

Anthropotomy demo notes. Source: OpenStax Anatomy and Physiology 2e, chapter 19 (CC BY 4.0).